Student ID:	a
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Desk No.:	

Practice Examination, Semester 2, 2025

Evolutionary Computation
COMP SCI 3316/7316

Course Coordinator: Mr Anthropic Claude

Total Duration: 180 mins

Questions	Time	Marks
Answer all 25 questions	180 mins	150 marks
		150 Total

Instructions for Candidates

- This is a Modified Closed-Book examination (refer to Materials)
- Write all answers within the space under the question on the exam paper
- If you need more space, use the extra blank pages at the end of the exam
- Write your desk number on each page
- Examination material must not be removed from the examination room

Materials permitted:

- Paper dictionary - general or translation
- One double sided A4 pages of notes

DO NOT COMMENCE WRITING UNTIL INSTRUCTED TO DO SO

STOP WRITING IMMEDIATELY WHEN INSTRUCTED

Part A: History and Foundations (20 marks)

Question 1: Evolutionary Computation History and Components [10 marks]

(a) Compare and contrast the four main approaches in evolutionary computation (Genetic Algorithms, Evolution Strategies, Evolutionary Programming, and Genetic Programming) by completing the following table. For each approach, specify:

- Traditional representation used
- Primary variation operator emphasized
- Typical selection method
- One typical problem domain

[6 marks]

(b) Explain why John Holland's original Genetic Algorithm emphasized crossover over mutation. Discuss the exploration vs. exploitation trade-off and explain what information each operator can uniquely provide to the search process.

[4 marks]

Part B: Genetic Algorithms and Representations (35 marks)

Question 2: Binary Representation Operations [12 marks]

Consider the following two parent bit strings of length $n = 12$:

Parent 1: `1 0 1 1 0 0 1 1 1 0 0 1`
Parent 2: `0 1 1 0 1 1 0 0 1 1 0 0`

(a) Perform **two-point crossover** with crossover points after positions 4 and 9. Show both offspring clearly indicating which segments come from which parent.

[3 marks]

(b) Perform **uniform crossover** on the original parents using the following random mask (1 = take from Parent 1, 0 = take from Parent 2):

Mask: `1 0 1 1 0 0 1 0 1 0 1 1`

Show both offspring.

[3 marks]

(c) Apply **bit-flip mutation** with mutation rate $p_m = 1/n$ to Offspring 1 from part (a). Assume bits at positions 3 and 11 are selected for flipping. Show the result.

[2 marks]

(d) Calculate the expected number of bits that will be flipped when applying bit-flip mutation with rate $p_m = 0.05$ to a population of 50 individuals, each of length $n = 100$ bits.

[4 marks]

Desk No.:

Question 3: Permutation Representation Operations [15 marks]

Consider the following two parent permutations representing TSP tours:

$$\text{Parent 1: } (3, 7, 1, 4, 6, 2, 5, 8)$$
$$\text{Parent 2: } (2, 4, 6, 1, 8, 7, 3, 5)$$

(a) Perform **Order Crossover (OX)** with the crossover segment from positions 3 to 6 (inclusive). Show your working step-by-step and clearly indicate both offspring.

[5 marks]

(b) Perform **Cycle Crossover (CX)** on the original parents. Identify all cycles and show both offspring.

[5 marks]

(c) Apply **Inversion Mutation** to Parent 1, inverting the segment from position 2 to position 5 (inclusive). Show the result.

[2 marks]

(d) Write pseudo-code for the Swap Mutation operator that swaps two randomly selected positions in a permutation.

[3 marks]

Desk No.:	

Question 4: Selection Methods [8 marks]

Consider a population of 10 individuals with the following fitness values (for a **maximization** problem):

Individual	A	B	C	D	E	F	G	H	I	J
Fitness	85	92	78	88	95	70	82	90	75	86

(a) Use **Fitness Proportionate Selection (Roulette Wheel)** to select 5 individuals. Calculate the selection probability for each individual and show your working. State which individuals would be selected if the random numbers generated are: 0.15, 0.42, 0.68, 0.83, 0.95.

[4 marks]

(b) Use **Tournament Selection** with tournament size $k = 3$ to select 5 individuals. For each tournament, assume the following individuals are randomly selected:

- Tournament 1: {B, F, H}
- Tournament 2: {A, D, I}
- Tournament 3: {C, E, J}
- Tournament 4: {B, G, I}
- Tournament 5: {D, F, H}

Show which individual wins each tournament and explain why.

[4 marks]

Desk No.:

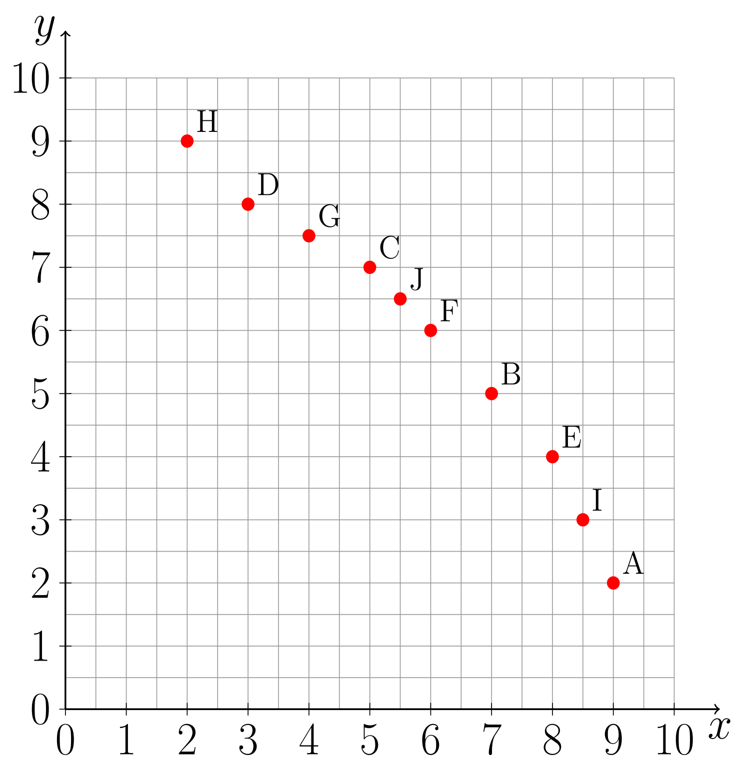
Part C: Evolutionary Multi-Objective Optimization (30 marks)

Question 5: NSGA-II Selection Procedure [15 marks]

Consider the following 10 solutions with their objective values for a **bi-objective maximization** problem (maximizing both f_1 and f_2):

Solution	f_1	f_2
A	9	2
B	7	5
C	5	7
D	3	8
E	8	4
F	6	6
G	4	7.5
H	2	9
I	8.5	3
J	5.5	6.5

(a) Draw all 10 objective vectors in a 2D objective space. Label each point clearly.



[2 marks]

(b) Perform **non-dominated sorting** to identify all fronts. For each solution, indicate:

- Which solutions it dominates
- Which solutions dominate it
- Its front number

Show your working clearly.

[6 marks]

(c) Calculate the **crowding distance** for all solutions in the first front. Use the boundary solutions and show all intermediate calculations.

[5 marks]

(d) Using NSGA-II environmental selection, select 5 solutions from the 10 to form the next generation. Justify your selection based on front numbers and crowding distances.

[2 marks]

Question 6: Hypervolume Indicator Selection [15 marks]

Consider the following 8 solutions for a **bi-objective maximization** problem with reference point $(0, 0)$:

Solution	f_1	f_2
P1	8	2
P2	7	4
P3	6	5
P4	5	6
P5	4	7
P6	3	7.5
P7	6.5	4.5
P8	5.5	5.5

(a) Draw all 8 solutions in the objective space. Identify which solutions are on the Pareto front (non-dominated).

[3 marks]

Desk No.:

(b) Calculate the total hypervolume dominated by all 8 solutions. Show your working by breaking down the area into rectangles.

[4 marks]

(c) Iteratively remove 3 solutions by always removing the solution with the smallest **hypervolume contribution** (the solution whose removal results in the smallest decrease in total hypervolume). For each iteration:

- Calculate each remaining solution's hypervolume contribution
- Identify which solution to remove
- Show the updated hypervolume

[8 marks]

Part D: Runtime Analysis of Evolutionary Algorithms (35 marks)

Question 7: Tail Inequalities [15 marks]

(a) State the conditions required for applying each of the following tail inequalities:

- Markov's Inequality
- Chebyshev's Inequality
- Cantelli's Inequality (One-sided Chebyshev)
- Chernoff Bounds

Discuss which inequality typically provides the tightest bound and why.

[5 marks]

Desk No.:

(b) A bit string of length $n = 400$ is chosen uniformly at random. Use **all four** tail inequalities (where applicable) to give upper bounds on the probability that the bit string has at least 250 ones.

For this problem:

- The expected number of ones is $\mu = 200$
- The variance is $\sigma^2 = 100$

Show all your calculations and compare the bounds obtained from each inequality.

[10 marks]

Question 8: Runtime Analysis of (1+1) EA on LeadingOnes [20 marks]

Consider the **LeadingOnes** function defined as:

$$\text{LeadingOnes}(x) = \sum_{i=1}^{n} \prod_{j=1}^{i} x_j$$

This function counts the number of leading ones in a bit string $x \in \{0,1\}^n$ and should be **maximized**.

(a) Explain what the LeadingOnes function computes. Give the LeadingOnes value for the following bit strings:

- $x_1 = (1, 1, 1, 0, 1, 0, 1, 1)$
- $x_2 = (1, 1, 1, 1, 1, 1, 1, 1)$
- $x_3 = (0, 1, 1, 1, 1, 1, 1, 1)$

[3 marks]

(b) Consider a search point x with $\text{LeadingOnes}(x) = i$ where $i < n$. What is the probability that the (1+1) EA creates an offspring y with $\text{LeadingOnes}(y) = i + 1$ in a single iteration? Explain your reasoning.

[4 marks]

Desk No.:

(c) Let T_i denote the expected number of iterations for the (1+1) EA to reach a search point with LeadingOnes value $i + 1$ starting from a search point with LeadingOnes value i. Show that:

$$T_i \leq \frac{en}{1} = en$$

[5 marks]

(d) Using the result from part (c), prove that the (1+1) EA obtains an optimal solution (all ones) in expected time $O(n^2)$ when started from an arbitrary search point.

[5 marks]

(e) Explain why LeadingOnes is considered harder for the (1+1) EA than OneMax. Compare their expected runtimes.

[3 marks]

 Desk No.:

Part E: Evolutionary Submodular Optimization (25 marks)

Question 9: Submodular Functions and Approximation [10 marks]

(a) Define what it means for a set function $f : 2^X \to \mathbb{R}$ to be:

- Submodular
- Monotone
- Normalized

Provide the mathematical definitions and explain the intuition behind submodularity (diminishing returns). [6 marks]

(b) Explain how a greedy algorithm can achieve a $(1 - 1/e)$-approximation for maximizing a monotone submodular function under a uniform cardinality constraint (selecting at most k elements). Provide the pseudo-code for this greedy algorithm.

[4 marks]

Question 10: Maximum Coverage Problem [15 marks]

Consider the graph $G = (V, E)$ where:

$$V = \{v_1, v_2, v_3, v_4, v_5, v_6\}$$

$$E = \{\{v_1, v_2\}, \{v_1, v_3\}, \{v_2, v_4\}, \{v_3, v_4\}, \{v_3, v_5\}, \{v_4, v_6\}, \{v_5, v_6\}\}$$

(a) Draw this graph clearly showing all vertices and edges.

[2 marks]

Desk No.:

(b) Consider the **bi-objective maximum coverage problem** where a bit string $x \in \{0,1\}^6$ represents which vertices are selected ($x_i = 1$ means vertex v_i is selected). The two objectives are:

- $f_1(x)$: Number of covered vertices (a vertex is covered if it is selected OR adjacent to a selected vertex)
- $f_2(x)$: $-|x|$ (negative of the number of selected vertices) to **minimize** selection

Find **all Pareto optimal solutions** and their objective vectors. Present your answer as a table showing each solution's bit string representation and its (f_1, f_2) values.

[8 marks]

(c) Draw the Pareto front in the objective space. How many Pareto optimal objective vectors are there?

[3 marks]

(d) Is the coverage function f_1 monotone and submodular? Justify your answer.

[2 marks]

Desk No.:	

Part F: Evolution Strategies (20 marks)

Question 11: Step Size Adaptation and Selection [20 marks]

(a) Explain why **step size adaptation** is crucial in Evolution Strategies. Discuss what happens if:

- The step size is too large
- The step size is too small
- The step size is not adapted during the search

[5 marks]

(b) Describe **three different methods** for adapting step sizes in Evolution Strategies:

- Self-adaptation
- 1/5 success rule
- Covariance Matrix Adaptation (CMA-ES) - brief overview

[6 marks]

(c) Consider the following population for a **maximization** problem:

Parents ($\mu = 3$):

- p_1: fitness = 85
- p_2: fitness = 78
- p_3: fitness = 92

Offspring ($\lambda = 5$):

- o_1: fitness = 88
- o_2: fitness = 75
- o_3: fitness = 95
- o_4: fitness = 82
- o_5: fitness = 90

Perform **(μ, λ)-selection** to select the next generation of 3 individuals. Explain your selection.

[3 marks]

(d) Using the same parent and offspring populations from part (c), perform **($\mu + \lambda$)-selection** to select the next generation of 3 individuals. Explain how this differs from (μ, λ)-selection.

[3 marks]

Desk No.:

(e) Write pseudo-code for (μ, λ)-selection that takes the offspring population as input and returns the selected individuals.

[3 marks]

Part G: Evolutionary Programming and Genetic Programming (15 marks)

Question 12: Finite State Machines and GP [15 marks]

(a) When evolving **Finite State Machines** using Evolutionary Programming, describe **three typical mutation operators** that can be applied. For each operator, explain what it does and provide a simple example.

[6 marks]

Desk No.:

(b) In Genetic Programming, solutions are traditionally represented as **tree structures**. Consider the following two parent trees representing mathematical expressions:

Parent 1: $(x + 3) \times (y - 2)$

Parent 2: $(x \times y) + (5 - x)$

- Draw both trees using standard GP tree notation (internal nodes are functions, leaves are terminals)
- Perform **subtree crossover** by swapping the subtree $(y - 2)$ from Parent 1 with the subtree $(5 - x)$ from Parent 2
- Show both offspring trees and their corresponding expressions

[5 marks]

(c) Explain what **bloat** means in Genetic Programming. Why is it a problem? Suggest **two methods** to control or prevent bloat.

[4 marks]

Desk No.:

Part H: Integration Question (20 marks)

Question 13: Multi-Objective LOTZ Problem [20 marks]

Consider the bi-objective function $\text{LOTZ} : \{0,1\}^n \to \mathbb{R}^2$ defined as:

$$\text{LOTZ}(x) = (\text{LO}(x), \text{TZ}(x))$$

where:

- $\text{LO}(x) = \sum_{i=1}^{n} \prod_{j=1}^{i} x_j$ (Leading Ones)
- $\text{TZ}(x) = \sum_{i=1}^{n} \prod_{j=n-i+1}^{n} (1 - x_j)$ (Trailing Zeros)

Both objectives should be **maximized**.

(a) For $n = 6$, compute $\text{LOTZ}(x)$ for the following bit strings:

- $x_1 = (1, 1, 1, 0, 0, 0)$
- $x_2 = (1, 1, 0, 0, 0, 0)$
- $x_3 = (1, 1, 1, 1, 0, 0)$
- $x_4 = (0, 0, 0, 0, 0, 0)$
- $x_5 = (1, 1, 1, 1, 1, 1)$

[5 marks]

(b) Determine the **complete set of Pareto optimal solutions** for $n = 6$. List all solutions and their objective vectors.

[5 marks]

(c) Draw the Pareto front in the objective space. What is the size of the Pareto front?

[2 marks]

Desk No.:

(d) Sketch a proof showing that GSEMO (Global Simple Evolutionary Multi-Objective Optimizer) obtains a population containing a solution for each Pareto optimal objective vector in expected time $O(n^3)$. You should:

- Identify the key phases of the search
- Bound the expected time for each phase
- Combine the bounds to get the overall result

[8 marks]

Extra Working Space

Use this space for additional working if needed. Clearly indicate which question your working relates to.